

# Group Theory Notes

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# Outline

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# Rubik's cube

Consider the Rubik's cube (for an online example, see [here](#)). What moves can we perform on a Rubik's cube?

## Rules of moves

There are many, *many* different configurations (or states) possible, but only one solution. We can get from one state to another by performing a valid movement of the cube. These movements have the following properties:

1. There is an unchanging list of allowable moves.
2. Every move is reversible.
3. Every move is deterministic.
4. Moves can be combined.

# Generators of Rubik's cube motions

The website above uses twelve basic moves:  $F$ ,  $F'$ ,  $U$ ,  $U'$ , and so on.

- How can you write  $F'$  in terms of  $F$ ?
- What is the smallest number of moves you can find that represent all of the basic moves on a Rubik's cube?

The observations above form the basis of a mathematical structure called a **group**. Groups appear throughout mathematics and its applications.

## Examples of groups

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## Coins in a line

Consider placing a penny and a nickel side by side in front of you. Is the action of swapping the positions of both coins a group? Note that you will need to specify an appropriate list of moves and determine if that list satisfies the above rules.

## Marbles in piles

Suppose you have five marbles in each of two piles in front of you. You have two actions available to you: you can take a marble from the left pile and move it to the right pile, *or* you can take a marble from the right pile and add it to the left pile. Is this a group?

## Three letter words

Consider all possible three letter "words" that can be formed from the letters  $\{A, B, C\}$ , where each letter must be used precisely once.

1. How many words are possible?
2. Consider two actions: swapping the first and second letter of a word, and swapping the second and third letter of a word. Can these two actions alone generate all of the words possible starting from  $ABC$ ?
3. Is this a group?

At this point, you might start thinking of a group as a set of objects that we can move around in a nice way. Group theory is about the study of such sets following the rules outlined above.

## Coins in a line (again)

Consider the group of coins in a line.

1. By rule 4, the action from this exercise when performed twice in a row yields another action in the group. What is this action?
2. Does every group have an action like this?

# Examples of groups

Try to find or construct groups that satisfy the following constraints:

1. The order in which the actions are performed matters.
2. The order in which the actions are performed doesn't matter.
3. There are exactly three actions.
4. There are exactly four actions.
5. There are infinitely many actions.

# Nonexamples of groups

Devise a situation that satisfies all of the group rules except for the second, third or fourth depending on your group.

# Rubik's cube group

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The **cubies** on a Rubik's cube can be placed in the following categories:

1. Corner cubies
2. Edge cubies
3. Center cubies

What happens to cubies in the previous categories if you apply any of the six basic Rubik's cube actions?

## Size of the Rubik's cube group

The Rubik's cube group is based on the six fundamental motions F, B, U, D, L and R. How large could this group be?